Curriculum Vitae

Abhishek Mishra

PhD researcher,
Laboratoire d'Information Quantique,
Université Libre de Bruxelles, Belgium.

Email: abhishek.mishra@ulb.be
abhisheksoloabh@gmail.com
Website: **https://abh1mishra.github.io**

Education

Oct 2022 – Cont.
Université Libre de Bruxelles

PhD in Quantum Information
Supervisor: Prof. Stefano Pironio

Aug 2017 – Jun 2022
Indian Institute of Science Education and Research, Kolkata

BS-MS Dual Degree , MS: 8.9/10 CGPA
Thesis title: *QKD security proofs and Device-Independent Certification.*
Thesis supervisors: Prof. Guruprasad Kar & Dr. Ramij Rahaman, ISI Kolkata.

Research Interests

Non-commutative Polynomial Optimization, Quantum Cryptography, Non-commutative Algebra

Skills

Julia, NumPy, SciPy, Python, C, Java, Origin, Probability Theory, Linear Algebra, Quantum Information, Convex Optimization, Semi-definite Programming, Mathematica, App Development, Blockchain.

Scientific Projects & Publications

1. Partially-commutative Polynomial Optimization

- Introduced a unified framework interpolating between commutative and non-commutative polynomial optimization, using Gröbner basis theory to significantly reduce the complexity of the optimization.
- Established foundations of partially-commutative Gröbner bases and derived explicit bases for a broad class of problems arising in quantum information.
- Developed efficient normal-form algorithm for trace polynomial optimization, superseding prior heuristic approaches.

Collaborators: Dr. Moisés Bermejo Morán, Dr. Stefano Pironio

Paper: To be published soon

2. PCPOP : A Julia library for polynomial optimization

- Implements normal-form algorithms introduced prior. User-friendly and versatile interface to solve a wide range of polynomial optimization problems with support of sparsity, symmetrization and built-in functions for QKD protocols.
- Efficient and general polynomial library support for Julia.
- Benchmarked against top libraries like OSCAR, QuantumNPA, Ncpol2sdpa etc. in terms of efficiency and mathematical rigour.

Collaborator: Dr. Moisés Bermejo Morán

Paper: To be published soon, Library Link

3. Semi-device Independent QKD

- Proposed secure QKD protocols using the restricted-information and restricted-distrust framework in prepare-and-measure scenarios; built numerical framework to demonstrate their robustness.
- Investigating efficient QKD protocols under the basis-independence assumption.

Collaborators: Dr. Michele Masini, Dr. Armin Tavakoli, Dr. Stefano Pironio, Dr. Chirag Srivastava

Paper: Two papers To be published soon.

4. Transmitter-Device-Independent QKD

- Proof-of-principle demonstration of 1Sided Device-Independent QKD over 140km.

Collaborators: Dr. Qiang Zeng

Paper: Sent to PRL

5. Device-Independent Certification (Part of master thesis)

- Proved robustness of self-testing of genuine multipartite entanglement.
- Built a library for solving NPA hierarchy for any Bell scenario.

Collaborator: Dr. Ramij Rahaman

Paper: Self-testing of genuine multipartite entangled states without network assistance

Contributed Talks & Posters

1. 828. WE-Heraeus-Seminar on Operator Theory and Polynomial Optimization in Quantum Information Theory, March 2025, Physikzentrum Bad Honnef

Talk : *Partially-commutative Polynomial Optimization*

2. Quantum Redemption, Aug 2024, Mehedeby

Talk : *Partially-commutative Polynomial Optimization*

3. VERIqTAS workshop, Mar 2026, Lyon [Accepted]

Talk : *Semi-Device-Independent Quantum Key Distribution Secure under Information and Distrust Assumptions*

4. POP23 – Future Trends in Polynomial Optimization, Nov 2023, LAAS-CNRS, Toulouse

Poster : *Partially commuting Computations*

5. QCRYPT 2024, Sep 2024, Vigo

Poster : *Partially commuting Computations*

6. IQOQI Summer School, Sep 2024, Vienna & Innsbruck

Poster : *Partially commuting Computations*

7. YQIS 2025, Oct 2025, ICFO, Barcelona

Poster : *Partially-commutative Polynomial Optimization*

Peer-review of Journals

Reviewer for international journals, including **Quantum** and Elsevier's **Conmputer Physics Communications**.

Research visits

Nov-Dec 2025	**Lund university, Sweden** Research visit in the group of Dr. Armin Tavakoli.
Jun 2025	**INRIA, Paris-Saclay** Research visit in the group of Dr. Marc-oliver Renou.
Aug 2022	**IIT Madras, Chennai** Project associate job in the Center For Quantum Information, Communication And Computing under Dr. Pradeep Sarvepalli.
Feb-Jul 2021	**ISI Kolkata** Research visit in the group of Prof. Guruprasad Kar and Dr. Ramij Rahaman.
Summer 2018	**IIT Kharagpur** Research visit in the group of Prof. Anushree Roy.
Summer 2019	**IIIT Hyderabad** Research visit in the group of Dr. Prabhakar Bhimalapuram.

Teaching Assistant

Spring 2022	**Quantum Information Processing** *Motivation and introduction, Quantum Algorithms and computation*

Extra-curricular Activities & Leadership

Web-Development	**Developed Websites or Apps for following:** Inquivesta 2019, Annual fest of IISER Kolkata CR and Admin portal of Inquivesta 2020 IICM 2021 (Part of it)
Initiatives	**Started the Coding and Designing Club of IISER Kolkata** Created the charter of the club. Formed the Council of the Office Bearers of the Club. Created the social media platform for the club.